

First, check whether SELinux is enabled on your system—as root, run the following command:

```
[root@vbg ~]# sestatus
```

If your output is:

```
SELinux status: disabled
```

...then you will have to enable SELinux on your server, as it is currently disabled.

Else, SELinux should be enabled in your system and the output will state so. We will explore the meaning of this output in the next part of this series.

To enable SELinux, edit the `/etc/sysconfig/selinux` file. Initiate SELinux in 'permissive' mode:

```
SELINUX=permissive.
```

And use the default targeted policy for SELinux Rules:

```
SELINUXTYPE=targeted
```

Once you set these values, save the file and reboot your system.

File relabelling

Once you have enabled SELinux in your system, you will notice that on the next reboot, all files in your system will be labelled. Labelling for files is akin to applying Extended Attributes required by SELinux to your ext2/ext3 filesystem. This process may take some time, so be patient and let the complete reboot occur.

Checking Security Contexts

Upon reboot, log in to your shell and issue the 'ls' command. Did you see any extended attributes? I doubt it.

To see the Security Context of files and directories, use `ls -Z`. You will see an output similar to the one below, containing the extended attributes:

```
[vbg@vbg selinux]$ ls -lZ
-rw-rw-r-- vbg vbg user_uobject_ruser_home_tso scratch-pad
-rw-rw-r-- vbg vbg user_uobject_ruser_home_tso series-definition
-rw-rw-r-- vbg vbg user_uobject_ruser_home_tso series-of-articles
-rw-rw-r-- vbg vbg user_uobject_ruser_home_tso The-Art-Of-Guard.vbg
```

You can now see the Security Context of all files/folders in your system. In the above snippet, for the file `scratch-pad`, the DAC related attributes are: owner(vbg), group(vbg), and permissions (664); while the MAC related Extended Attributes (xattrs) that define the Security Context are: `user_uobject_ruser_home_tso`. The fourth attribute in the Security Context defines the sensitivity. We will discuss this fourth attribute in the later part of the series.

To check the ID of the logged in user, issue the

command `id -Z`:

```
[vbg@vbg selinux]$ id -Z
user_u.system_runconfined_tso
```

The `id` command also gives the UID/GID of the user:

```
[vbg@vbg selinux]$ id
uid=500(vbg) gid=500(vbg) groups=500(vbg) context=user_u.system_r:
unconfined_tso
```

To see a list of processes (subjects) with their security contexts, issue the `ps auxZ` command*:

```
[vbg@vbg selinux]$ ps auxZ
user_u.system_runconfined_tso 3543 ? Ss 0:01 /usr/bin/perl -T -w
/usr/bin/spamd -s socketpath /home/vbg/.evolu
user_u.system_runconfined_tso 3546 ? S 0:02 spamd child
user_u.system_runconfined_tso 3730 pts/3 Ss 0:00 /bin/bash
user_u.system_runconfined_tso 3760 ? S 0:00 notify [kdeinit]
user_u.system_runconfined_tso 3762 pts/3 S+ 0:02 vim The-Art-Of-
Guard.vbg
user_u.system_r.htpd_tso root 4227 41.0 1.4 42376 29216 pts/2
R+ 14:46 0:01 /usr/sbin/htpd
```

Please note that each process has been assigned a 'security_context'. Do also note the type assigned to these subjects.

Try out the above introductory commands in SELinux. Explore... there is a wealth of information available on SELinux on the Web. Do not enforce a Strict Policy while you are not too sure about SELinux Policies, which we will cover in detail in the coming issues. If you end up in a state that your system will not boot satisfactorily or if you would simply like to disable SELinux at boot time, press `e` on your GRUB screen to edit the boot sequence when rebooting. Edit the default kernel booting parameters by appending `"selinux=0"` at the end of the line and continue to boot.

This will temporarily disable SELinux and you will be back to traditional DAC security.

Coming up

- Enforcing and Permissive Modes
- Controlling SELinux
- Targeted Policy
- Understanding the Targeted Policy
- Policy Modules 

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